

Chapter 51

Animal Behavior

PowerPoint® Lecture Presentations for

Biology

Eighth Edition

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Overview: Shall We Dance?

- Animal behavior is based on physiological systems and processes.
- A **behavior** is the *nervous system's response to a stimulus* and is carried out by the *muscular* or the *hormonal* system.
- Behavior helps an animal
 - Obtain food
 - Find a partner for sexual reproduction
 - Maintain homeostasis.

Animal Behavior is subject to Natural Selection.

- An animal's behavior is its response to external and internal stimuli.
- ***Ethology*** is the scientific *study of animal behavior*, particularly in natural environments.
- According to early ethologist Niko **Tinbergen**, four questions should be asked about behavior:
 1. What stimulus elicits the behavior, and what physiological mechanisms mediate the response?
 2. How does the animal's experience during growth and development influence the response mechanisms?

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3. How does the **behavior** aid survival and reproduction? **Survival value**
 4. What is the behavior's evolutionary history?
 3. These questions highlight the complementary nature of proximate and ultimate perspectives.
 4. **Behavioral ecology** is the study of the **ecological** and **evolutionary** basis for **animal behavior**.
 5. It integrates proximate and ultimate explanations for animal behavior.

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- **Proximate causation**, or “how” explanations, focus on
 - Environmental stimuli that trigger a behavior
 - Genetic, physiological, and anatomical mechanisms underlying a behavior.
 - **Ultimate causation**, or “why” explanations, focus on
 - Evolutionary significance of a behavior.

Fixed Action Patterns *FAP*

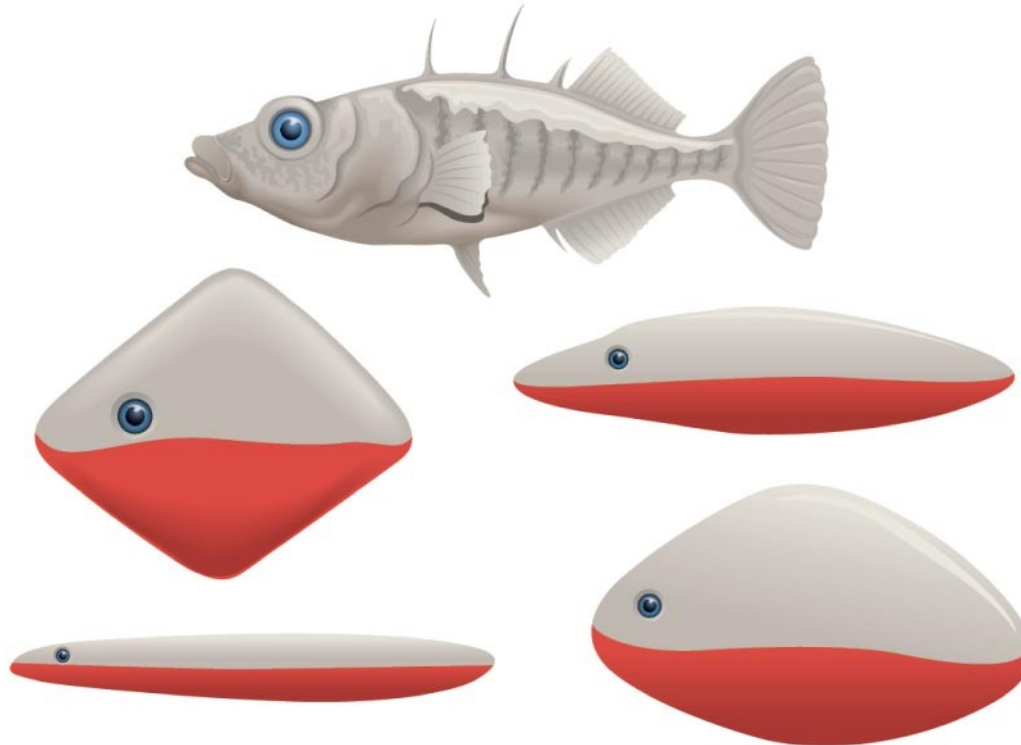
- A **fixed action pattern** is a sequence of unlearned, **innate** behaviors that is **unchangeable**.
- *Once initiated*, it is usually *carried to completion*.
- A fixed action pattern is triggered by an **external cue** known as a **sign stimulus**.

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- In male stickleback fish, the stimulus for attack behavior is the red underside of an intruder.
 - When presented with unrealistic models, as long as some red is present, the attack behavior occurs.

Sign stimuli in a
classic **FAP**
fixed action
pattern



(a)



(b)

Oriented / Directional Movement = Taxis ...

- **Environmental cues** can trigger movement in a particular direction.
- A **taxis** is a more or less automatic, oriented movement **toward or away from a stimulus**.
- Many stream fish exhibit a positive taxis and automatically swim in an upstream direction.
- This taxis prevents them from being swept away and keeps them facing the direction from which food will come.

Kinesis: non-directional change ...

- A **kinesis** is a simple **change** in activity or turning **rate** in response to a **stimulus**.
- For example, sow bugs become more active in dry areas and less active in humid areas.
- Though sow bug behavior varies with humidity, sow bugs do not move toward or away from specific moisture levels.

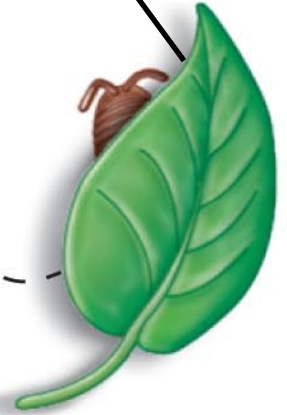
Kinesis - Sow bugs become more active in dry areas and less active in humid areas

Dry open area



Sow bug

Moist site under leaf



Migration

- **Migration** is a *regular, long-distance change in location*.
- **Animals** can **orient** themselves using
 - The position of the **sun** and their **circadian clock**, an internal 24-hour clock that is an integral part of their nervous system
 - The position of the **North Star**
 - The Earth's **magnetic field**.

Oriented Movement / seasonal: Migration



Behavioral Rhythms

- Some animal behavior is affected by the animal's *circadian rhythm*, a *daily* cycle of rest and activity.
- Behaviors such as *migration* and *reproduction* are linked to changing *seasons*, or a *circannual rhythm*.
- Some behaviors are linked to *lunar cycles*
 - For example, courtship in fiddler crabs occurs during the new and full *moon*.

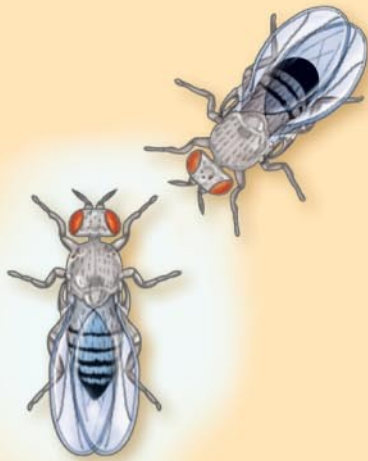
Male fiddler crab beckoning to potential mates



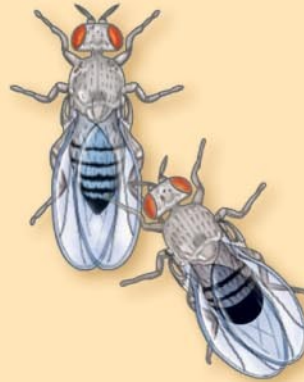
Animal Signals and Communication

- In behavioral ecology, a **signal** is a behavior that causes a change in another animal's behavior.
- **Communication** is the transmission and reception of signals.
- **Animals communicate** using visual, chemical, tactile, and auditory signals.
- The type of signal is closely related to lifestyle and environment.

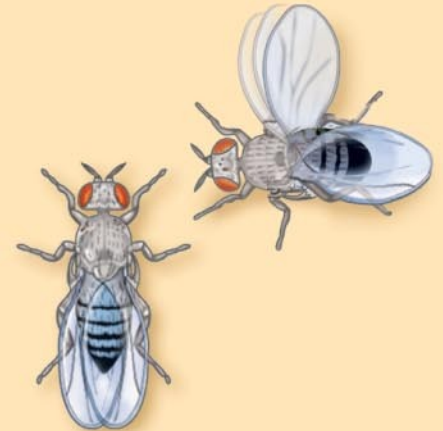
Courtship behavior of the fruit fly



(a) Orienting



(b) Tapping



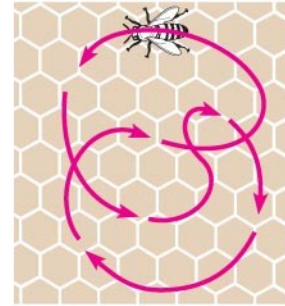
(c) “Singing”

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- **Honeybees** show complex communication with symbolic language.
 - A **bee** returning from the field performs a **dance** to communicate information about the position of a **food source**.

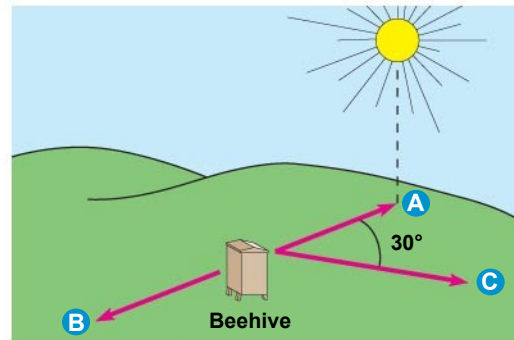
Honeybee dance language



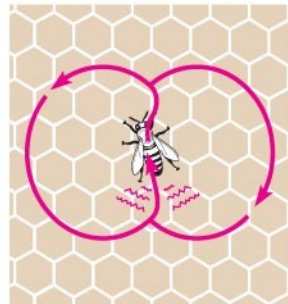
(a) Worker bees



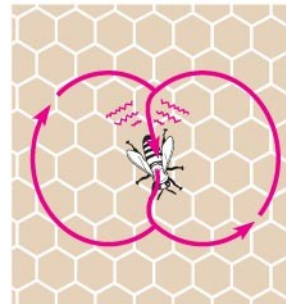
(b) Round dance
(food near)



(c) Waggle dance
(food distant)



Location **A**



Location **B**

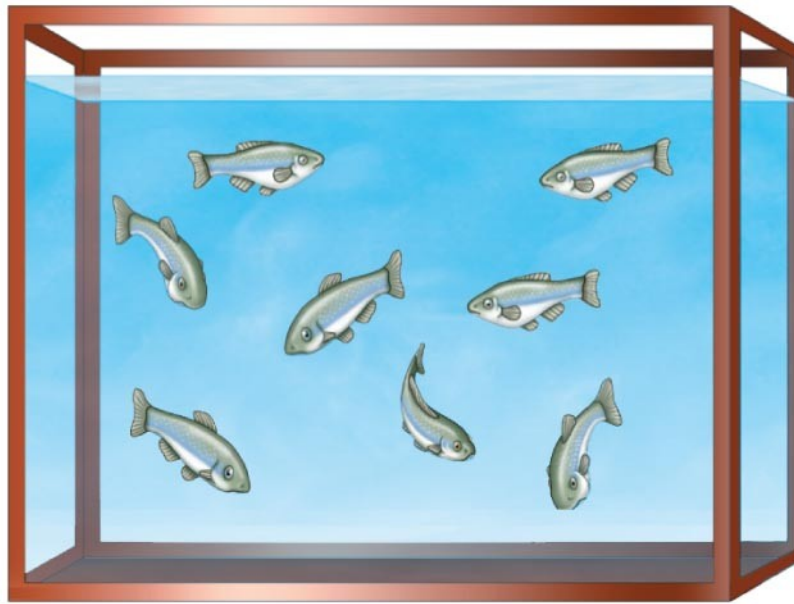


Location **C**

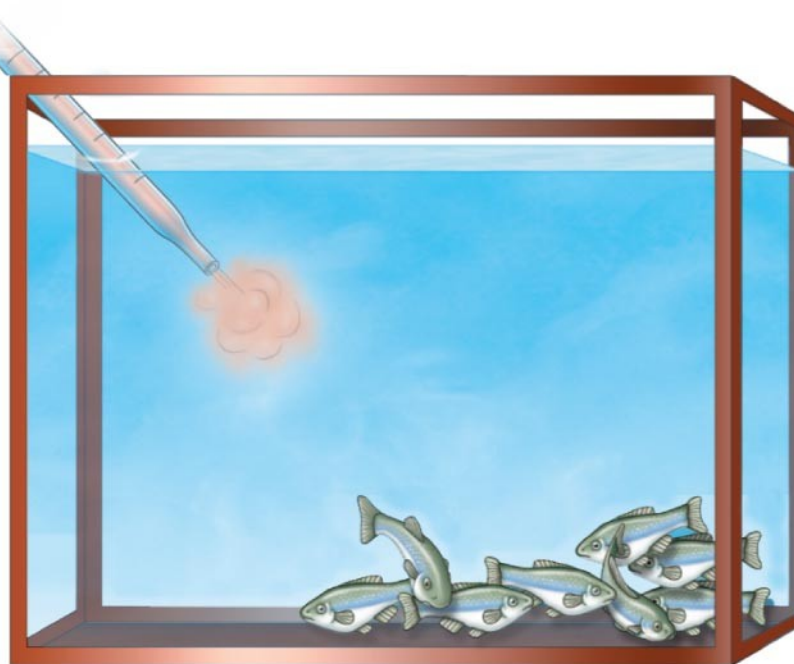
Pheromones

- Many **animals** that **communicate** through odors emit **chemical** substances called ***pheromones***.
- Pheromones are effective at very low concentrations.
- When a minnow or catfish is injured, an alarm substance in the fish's skin disperses in the water, inducing a fright response among fish in the area.

*Minnows
responding
to the
presence
of an
alarm
substance*



**(a) Minnows
before
alarm**



**(b) Minnows
After **pheromone**
alarm signal**

Learning establishes specific **links** between **experience** and **behavior**

- **Innate behavior** is developmentally **fixed** and under strong genetic influence / inborn.
- **Learning** is the **modification of behavior** based on specific **experiences**.

Habituation

- **Habituation** is a simple form of **learning** that involves **loss** of **responsiveness** to **stimuli** that convey little or no information. *Stop attending to a stimulus that is irrelevant.*
 - For example, birds will stop responding to alarm calls from their species if these are not followed by an actual attack.

Imprinting

- **Imprinting** is a behavior that includes a specific **critical period learning** and innate components and is generally **irreversible**.
- It is distinguished from other learning by a **sensitive period**.
- A sensitive period is a limited **developmental phase** that is the **only time** when certain behaviors can be learned.

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- An example of imprinting is young geese following their mother.
 - Konrad Lorenz showed that when baby geese spent the first few hours of their life with him, they imprinted on him as their parent.
 - Conservation biologists have taken advantage of imprinting in programs to save the whooping crane from extinction. Young whooping cranes can imprint on humans in “crane suits” who then lead crane migrations using small aircraft.

Imprinting



(a) Konrad Lorenz and geese



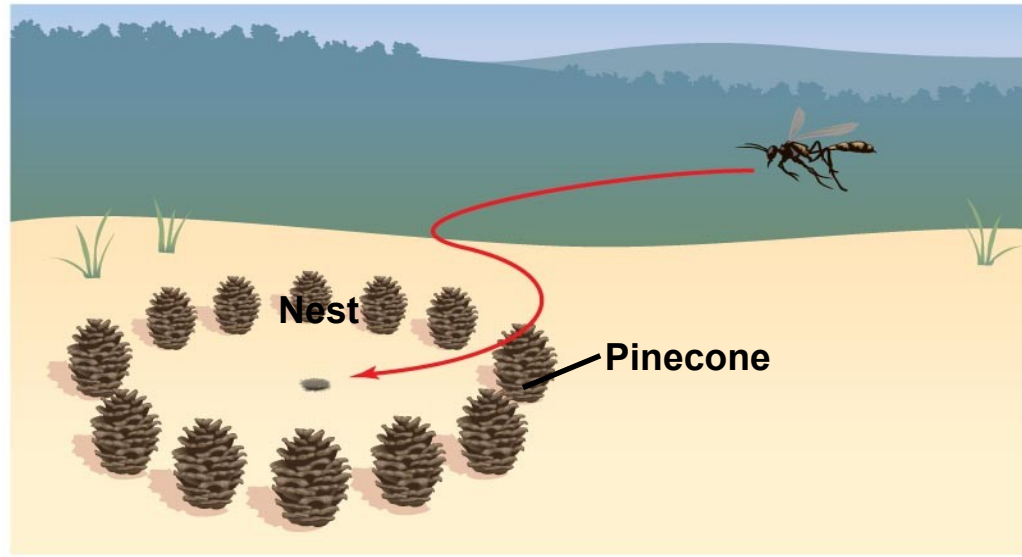
(b) Pilot and cranes

Spatial Learning

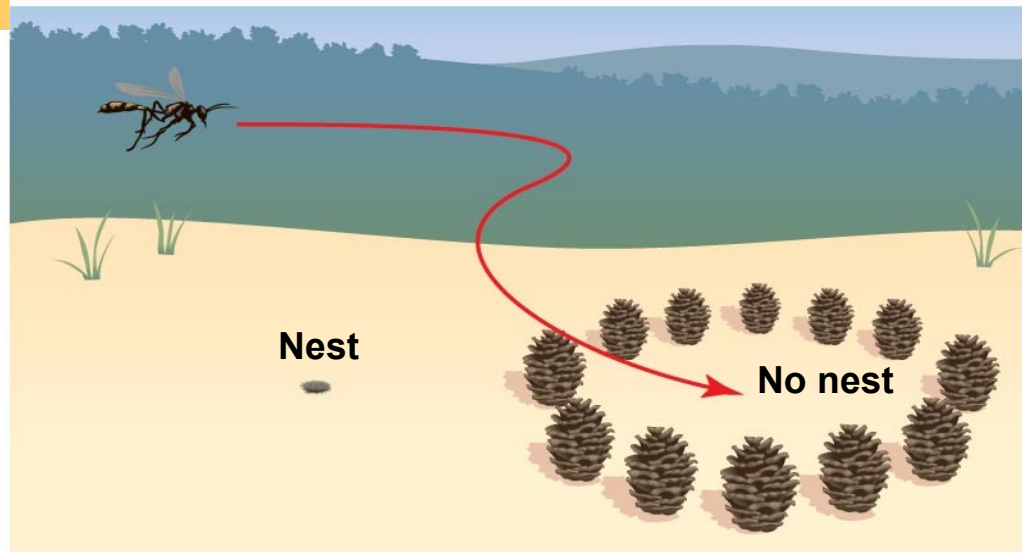
- **Spatial learning** is a more complex modification of behavior based on experience with the spatial structure of the environment.
- Niko Tinbergen showed how digger wasps use **landmarks** to find nest entrances.
- A **cognitive map** is an internal representation of spatial relationships between objects in an animal's surroundings often using particular landmarks.

Does a digger wasp use landmarks to find her nest?

EXPERIMENT



RESULTS



Associative Learning

- In **associative learning**, animals associate one feature of their environment with another.
Example: a mouse will avoid eating caterpillars with specific colors after a bad experience with a distasteful monarch butterfly caterpillar.
- **Classical conditioning** is a type of associative learning in which an arbitrary stimulus is associated with a reward or punishment.
Example: a dog that repeatedly hears a bell before being fed will salivate in anticipation at the bell's sound.

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- **Operant conditioning** is a type of **associative learning** in which an animal learns to associate one of its behaviors with a reward or punishment.
 - It is also called **trial-and-error learning**.

Example: a rat that is fed after pushing a lever will learn to push the lever in order to receive food.

Example: a predator may learn to avoid a specific type of prey associated with a painful experience.

Cognition and Problem Solving

- **Cognition** is a **process of knowing** that may include awareness, reasoning, recollection, and judgment.
 - For example, honeybees can distinguish “same” from “different.”

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- **Problem solving** is the process of devising a strategy to overcome an obstacle.

Example: chimpanzees can stack boxes in order to reach suspended food.

- Some animals learn to solve problems by observing other individuals.

Example: young chimpanzees learn to crack palm nuts with stones by copying older chimpanzees

A young chimpanzee learning to crack oil palm nuts by observing an experienced elder



Development of Learned Behaviors

- Development of some behaviors occurs in **distinct stages**.
 - For example a white-crowned sparrow memorizes the song of its species during an early sensitive period.
 - The bird then learns to sing the song during a second learning phase.

Both genetic makeup and environment contribute to the development of behaviors

- **Animal behavior** is governed by complex interactions between genetic and environmental factors.
- Cross-fostering studies help behavioral ecologists to identify the contribution of environment to an animal's behavior.
A **cross-fostering study** places the young from one species in the care of adults from another species.

Regulatory Genes and Behavior

- A master regulatory gene can control many behaviors.

Example: a single gene controls many behaviors of the male fruit fly courtship ritual.

- Multiple independent genes can contribute to a single behavior.

Example: in green lacewings, the courtship song is unique to each species; multiple independent genes govern different components of the courtship song.

Genetically Based Behavioral Variation in Natural Populations

- When **behavioral variation** within a species **corresponds** to **environmental variation**, it may be evidence of **past evolution**.

Case Study: *Variation in Migratory Patterns*

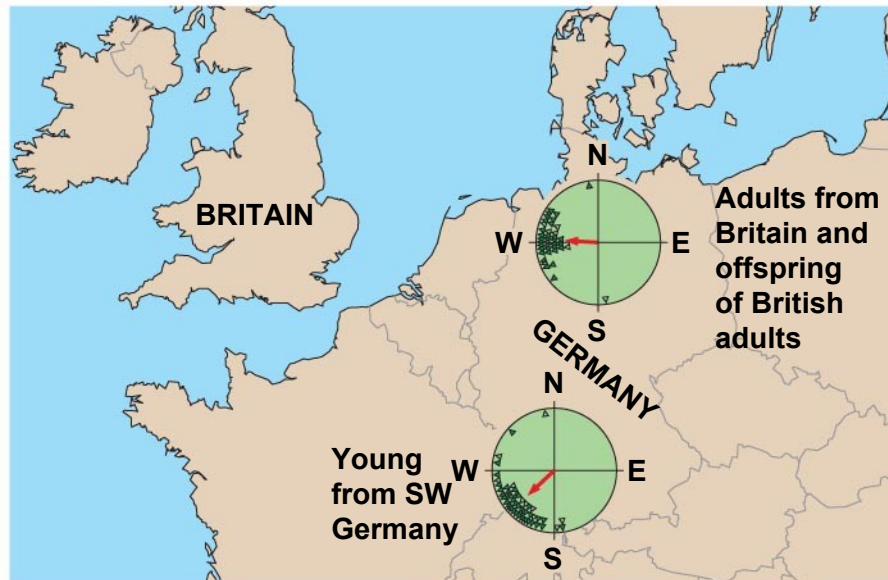
- Most blackcaps (birds) that breed in Germany winter in Africa, but some winter in Britain.
- The two migratory populations are genetically distinct.

Are differences in migratory orientation within a species genetically determined?

EXPERIMENT



RESULTS



Case Study: *Variation in Prey Selection*

- The **natural diet** of western garter snakes **varies by population**.
- Coastal populations feed mostly on banana slugs, while inland populations rarely eat banana slugs.
- Studies have shown that the **differences** in diet are **genetic**.
- The two populations **differ** in their **ability to detect** and **respond** to **specific odor molecules** produced by the banana slugs.

Western garter snake from a coastal habitat eating a banana slug



Natural Selection for individual Survival and Reproductive Success can explain most behaviors

- *Genetic components of behavior evolve through natural selection.*
- Behavior can affect **fitness** by influencing foraging and mate choice.
- **Natural selection** refines behaviors that enhance the efficiency of feeding.
- **Foraging** = food-obtaining behavior. Foraging includes recognizing, searching for, capturing, and eating food items.

Optimal Foraging Model

- **Optimal foraging model** views foraging behavior as a **compromise** between **benefits** of nutrition and **costs** of obtaining food.
- The costs of obtaining food include energy expenditure and the risk of being eaten while foraging.
- ***Natural selection*** should *favor foraging behavior that minimizes the costs and maximizes the benefits.*

Balancing Risk and Reward

- Risk of predation affects foraging behavior.
 - For example, mule deer are more likely to feed in open forested areas where they are less likely to be killed by mountain lions.

Mating Behavior and Mate Choice

- Mating behavior includes seeking or attracting mates, choosing among potential mates, and competing for mates.
- **Mating behavior** results from a type of natural selection called **sexual selection**.
- The mating relationship between males and females varies greatly from species to species.
- In many species, mating is **promiscuous**, with no strong pair-bonds or lasting relationships.

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- In **monogamous** relationships, **one male mates with one female**.
 - Males and females with monogamous mating systems have similar external morphologies.

*Relationship
between **mating
system**
and
**male and female
forms.***



(a) Monogamous species



(b) Polygynous species



(c) Polyandrous species

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- In **polygamous relationships**, an individual of one sex mates with several individuals of the other sex.
 - Species with polygamous mating systems are usually **sexually dimorphic**: males and females have different external morphologies.
 - **Polygamous relationships** can be either **polygynous** or **polyandrous**. In **polygyny** - *one male mates with many females*. The males are usually more showy and larger than the females.

Fig. 51-20b



Polygynous species – Male larger and more dominant

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- In **polyandry** = one female mates with many males.
 - The females are often more showy than the males.
 - Polyandry is a rare mating system.

Fig. 51-20c



Polyandrous species – female has multiple partners

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- *Needs of the young are an important factor constraining evolution of mating systems.*
 - Consider bird species where chicks need a continuous supply of food.
 - A male maximizes his reproductive success by staying with his mate, and caring for his chicks (monogamy).

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- Consider bird species where chicks are soon able to feed and care for themselves
 - A male maximizes his reproductive success by seeking additional mates (polygyny).
 - Females can be certain that eggs laid or young born contain her genes; however, paternal certainty depends on mating behavior.
 - *Certainty of paternity influences parental care and mating behavior.*

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- Paternal certainty is relatively low in species with internal fertilization because mating and birth are separated over time.
 - Certainty of paternity is much higher when egg laying and mating occur together, as in external fertilization.
 - *In species with external fertilization, parental care is at least as likely to be by males as by females.*

Paternal care by a male jawfish



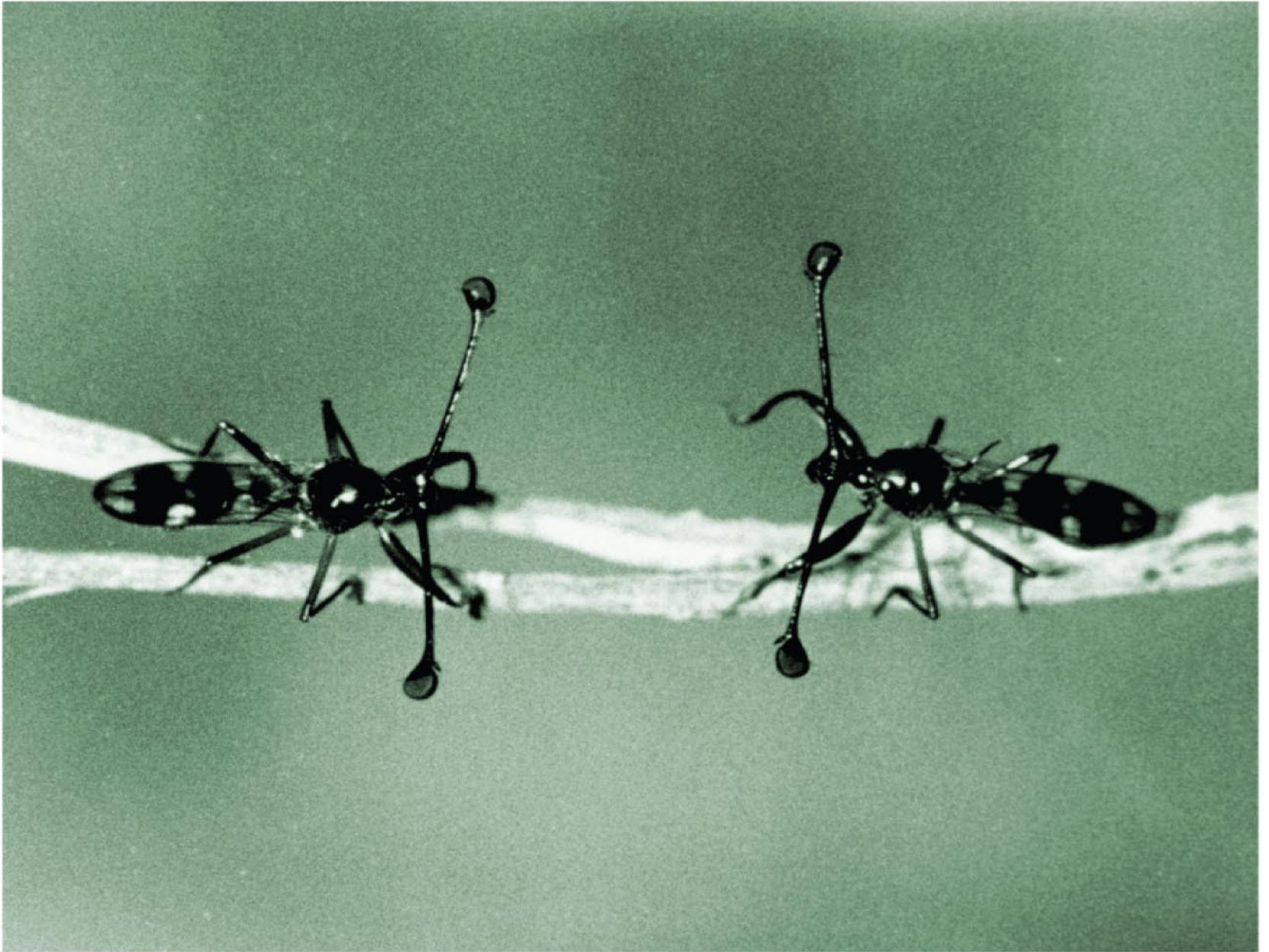
Sexual Selection and Mate Choice

- In intersexual selection, members of *one sex choose mates on the basis of certain traits.*
- Intrasexual selection involves *competition between members of the same sex for mates.*

Mate Choice by Females

- ***Female choice*** is a type of *intersexual competition*.
- Females can drive sexual selection by choosing males with **specific behaviors** or **features** of anatomy.
- For example, female stalk-eyed flies choose males with relatively long eyestalks.
- Ornaments, such as long eyestalks, often correlate with health and vitality.

Female choice - Male stalk-eyed flies



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- Another example of mate choice by females occurs in zebra finches.
 - *Female chicks who imprint on ornamented fathers are more likely to select ornamented mates.*
 - Experiments suggest that mate choice by female zebra finches has played a key role in the evolution of ornamentation in male zebra finches.

Appearance / variation of male zebra finches in nature



Male Competition for Mates

- Male competition for mates is a source of ***intrasexual*** selection that can reduce variation among males.
- Such competition may involve **agonistic behavior**, an often **ritualized contest** that determines which competitor gains access to a resource.

Males Compete - Agonistic interaction



Applying Game Theory

- In some species, sexual selection has driven the evolution of alternative mating behavior and morphology in males.
- The fitness of a particular phenotype (behavior or morphology) depends on the phenotypes of other individuals in the population.
- **Game theory** evaluates **alternative strategies** where the **outcome** depends on each **individual's strategy** and the strategy of other individuals.

-
- For example, each side-blotched lizard has a blue, orange, or yellow throat, and each color is associated with a **specific strategy for obtaining mates**. There is a genetic basis to throat color and mating strategy.
 - Like rock-paper-scissors, **each strategy will outcompete one strategy, but be outcompeted by the other strategy**. The success of each strategy depends on the frequency of all of the strategies; this **drives frequency-dependent selection**.

Male polymorphism in the side-blotched lizard (*Uta stansburiana*)



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Inclusive fitness can account for the evolution of altruistic social behavior

- Natural selection favors behavior that maximizes an individual's survival and reproduction.
- These behaviors are often selfish.
- On occasion, some **animals behave** in ways that **reduce their individual fitness** but **increase the fitness of others**.
- This kind of behavior is called **altruism**, or **selflessness**.

Altruism

Example of altruism / selfless behavior for the good of the group:

- Under threat from a predator, an individual Belding's ground squirrel will make an **alarm call to warn others**, even though calling increases the chances that the caller is killed.

Inclusive Fitness

- Altruism can be explained by inclusive fitness.
- **Inclusive fitness** is the **total effect** an individual has on proliferating its genes by producing offspring *and* helping close relatives produce offspring.

Hamilton's Rule and Kin Selection

- William Hamilton proposed *a quantitative measure for predicting when natural selection would favor altruistic acts among related individuals*.
- Three key variables in an altruistic act:
 - **Benefit** to the recipient (B)
 - **Cost** to the altruist (C)
 - **Coefficient of relatedness** (the fraction of genes that, on average, are shared; r)

-
- *Natural selection favors altruism when:*

$$rB > C$$

- This inequality is called **Hamilton's rule**.
- **Kin selection** is the natural selection that favors this kind of altruistic behavior by enhancing reproductive success of relatives.

Reciprocal Altruism

- Altruistic behavior toward **unrelated individuals** can be adaptive if the aided individual **returns the favor in the future**. This type of altruism is called **reciprocal altruism**.
- Reciprocal altruism is limited to species with **stable social groups** where individuals meet repeatedly, and cheaters (don't reciprocate) are punished. **Reciprocal altruism** has been used to explain altruism between **unrelated individuals in humans**.

Social Learning

- **Social learning** is learning through the observation of others and forms the roots of culture.
- **Culture** is a system of information transfer through observation or teaching that influences behavior of individuals in a population.
- Culture can alter behavior and influence the fitness of individuals.

Case Study: *Mate-Choice Copying*

- In **mate-choice copying**, individuals in a population copy the mate choice of others.
- This type of behavior has been extensively studied in the guppy *Poecilia reticulata*.
- Females who mate with males that are attractive to other females are more likely to have sons that are attractive to other females.

Case Study: *Social Learning of Alarm Calls*

- Vervet monkeys produce distinct alarm calls for different predators.
- Infant monkeys give indiscriminating calls but learn to fine-tune them by the time they are adults.

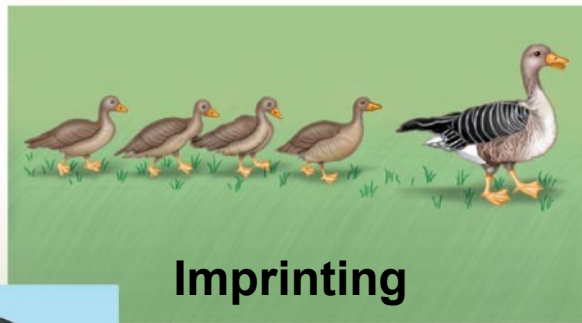
Vervet monkeys learning correct use of alarm calls



Evolution and Human Culture

- No other species comes close to matching the **social learning** and **cultural transmission** that occurs among humans.
- **Human culture** is related to evolutionary theory in the distinct discipline of **sociobiology**. Human behavior, like that of other species, results from **interactions** between **genes** and **environment**.
- *However, our social and **cultural institutions** may provide the only feature in which there is **no continuum between humans and other animals**.*

Learning

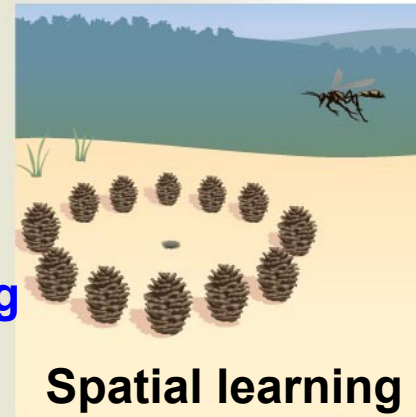


Imprinting



Cognition

**Learning
and
problem solving**



Spatial learning



Associative learning



Social learning

You should now be able to:

1. State Tinbergen's four questions and identify each as a proximate or ultimate causation.
2. Distinguish between the following pairs of terms: kinesis and taxis, circadian and circannual behavioral rhythms, classical and operant conditioning.
3. Suggest a proximate and an ultimate cause for imprinting in newly hatched geese.
4. Explain how associative learning may help a predator avoid toxic prey.

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5. Describe how cross-fostering experiments help identify the relative importance of environmental and genetic factors in determining specific behaviors.
 6. Describe optimal foraging theory.
 7. Define and distinguish among promiscuous, monogamous, and polygamous mating systems.
 8. Distinguish between intersexual and intrasexual selection.

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9. Explain how game theory may be used to evaluate alternative behavioral strategies.
 10. Define altruistic behavior.
 11. Distinguish between kin selection and reciprocal altruism.
 12. Define social learning and culture.